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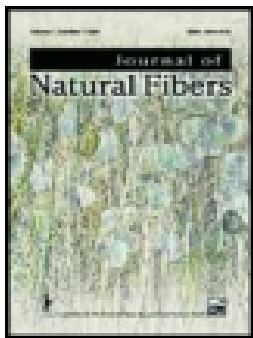


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Advancements in Applications of Natural Wool Fiber: Review

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ABSTRACT

Natural fibers are getting attention from researchers and academicians for utilization in new technology due to their ecofriendly nature and sustainability. This paper reviews advancements of utilization of natural wool fiber in industry and agriculture. Sheep wool is one of the natural and renewable sources which is widely used in the textile industry to make clothes, rugs, socks, and carpets, besides some uses in agriculture as fertilizers. Recently, wool has become alternative material in some industrial sectors, which increases the value of wool fiber. Increasing demand for alternative materials in buildings sectors has driven the production of sheep wool fibers with comparable and unique properties, for example thermal and sound insulation. Wool with enlarged pores is commercially available for sorption of dyes, oil spill, and volatile organic compounds (Greener Cleaner). Wool is also chemically modified for use in catalytic converter. The unique composition of wool has made it favorable in construction of light aircraft, ships, and trains upholstery. The self-extinguish property makes wool promising as a seal insulator for heat-generating appliances. Substances such as lanoline, ceramide, peptone, and keratin have been proposed for manipulation with wool to be utilized in pharmacotherapy cosmetics, and medical treatment.

KEYWORDS

Sheep wool; wool advancements; wool's substances. application of wool

关键词

羊羊毛; 羊毛进步; 羊毛的物质; 应用程序的羊毛

摘要

天然纤维具有良好的生态友好性和可持续性,在新技术领域的应用越来越受到研究者和学者的重视。本文综述了天然羊毛纤维在工农业上的应用进展。绵羊毛是一种天然的可再生资源,广泛应用于纺织工业,用于制造布料、地毯、袜子和地毯,还可用于农业,用作肥料。近年来,羊毛已成为一些工业部门的替代材料,这增加了羊毛纤维的价值。建筑部门对替代材料的需求不断增加,推动了具有可比和独特性能的羊毛纤维的生产,例如隔热和隔音材料。毛孔粗大的羊毛可用于吸附染料、溢油和挥发性有机化合物(更环保的清洁剂)。羊毛也经过化学改性用于催化转化器。羊毛的独特成分使其成为制造轻型飞机、轮船和火车的装饰材料。羊毛的自熄特性使其有望成为热发电设备的密封绝缘体。有人提出将羊毛中的羊毛碱、神经酰胺、蛋白胍和角蛋白等物质用于药物治疗、化妆品和医疗。

Introduction

Wool is an environmentally friendly fiber (Russell 2009). The structure and chemical composition of wool fiber differ it from other kinds of fibers; its huge diversity, heterogeneity of characteristics, and advantages are like no other natural or manmade fiber (Laitala, Klepp, and Henry 2018). Use of wool in apparel and textile manufacturing was established in the ancient era. However, wool has not been used to its full potential. There is a growing interest for the production of new composite materials to replace synthetic fibers with natural fibers in technical applications. Global sheep population

growth is projected to increase from 1.7 billion to 2.7 billion between 2000 and 2050 (Thornton 2010). Global sheep population in 2010 was 2 billion, and by 2020, is expected to reach around 2.3 billion in the 'reference world CWANA, Central and West Asia and North Africa; ESAP, East and South Asia and the Pacific; LAC, Latin America and the Caribbean; NAE, North America and Europe; SSA, sub-Saharan Africa (Rosegrant et al. 2009). Researchers are progressively finding new approaches to use natural wool instead of synthetic ones, which is environmentally friendly and at the lowest cost.

Wool is composed of single protein called keratin, which contains main five elements (in %) hydrogen 6–7 carbon 50, oxygen 21–24, nitrogen 15–21, sulfur 2–5, and other elements. Chemical features of wool include high content of pure keratin that contains amino acids, such as Glycine, Alanine, Serine, Proline, Naline, Threonine, Cystine, Lucien isomers, Aspartic acid Glutamic acid, Methionine, Histidine, Hydroxylsine, Phenylalanine, Arginin, Tyrosine, and Tryptophane. Wool is a copolymer of 18 amino acids, whereas synthetic fibers are copolymers with two monomers. Different content of amino acids in wool fiber contribute to its advantages of chemical characterization. Cystine containing sulfur is an important wool fiber property. Higher sulfur amount in wool means better treating properties, higher resistance to chemical impacts, and higher physicomachanical properties (Pekhtasheva, 2012). Wool fiber layers are composed of two types of cell: the internal cells of the cortex and external cuticle cells that form a sheath around the fiber, as shown in Figure 1. Cuticle cells (or scales), which overlap like tiles on a roof, make wool unique amongst textile fibers. An important function of cuticle cells is to anchor wool fibers in the skin of sheep. The exposed edge of each cuticle cell points from the fiber root toward the tip. This gives rise to a larger surface frictional value when a fiber is drawn in the against-scale direction, than in the with-scale direction. The frictional difference helps to expel dirt and other contaminants from the fleece, which is also responsible for wool's property of felting when agitated in water. This characteristic, which is not shared by any other textile fiber, enables fabrics with very dense structures to be produced, such as blankets, felts, and overcoat materials. Felting is generally undesirable, for example, in knitted garments that are machine-washed, but now, methods are available to remove the frictional difference and make wool shrink resistant. The fiber surface is also largely responsible for the natural softness of wool, which makes it one of the smoothest textile fibers.

Thermal energy storage (TES) plays an important role in optimizing thermal processes, helping to reduce the mismatch between the energy demand and supply, taking advantage of waste heat that otherwise would be lost, giving opportunity to implement peak load shifting strategies, and allowing better integration of renewable energies. However, the implementation

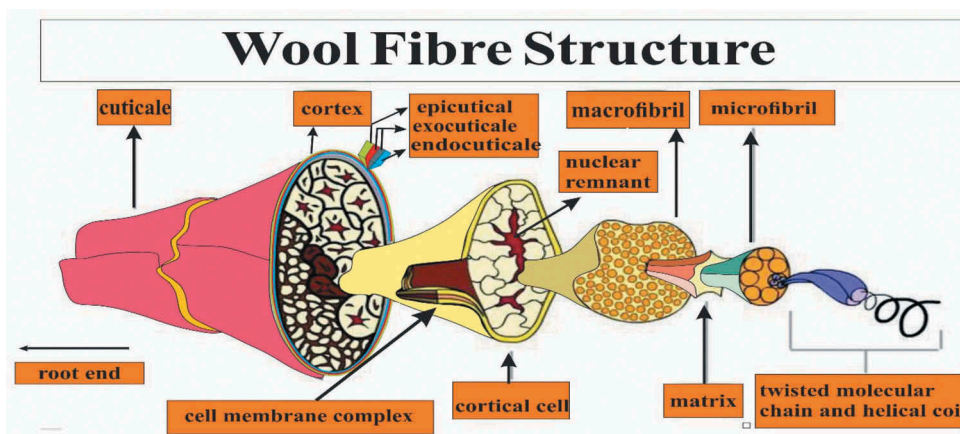


Figure 1. Wool structure.

cost of TES system is one of its main drawbacks, thus the cheap and commercially abundant wool is more favorable as enhancement in order to overcome the abovementioned drawback due to economic and easy-to-implement aspects.

Applications of sheep wool

Wool fiber is widely used in technical applications that can make use of its genetic characterization (Johnson 2003). Wool has been used in different fields, in which its unique properties and opportunities for specific enhancement are profitable (Fangueiro 2009). Wool has long been used in apparel and textile industry since the ancient era, and will contribute more in wide range of new and future utilizations. Still, it is perhaps lesser known that wool is increasingly utilized in technical applications nowadays (McNeil 2017). This drives the desire among wool producers to increase production, value of their fiber, and the recognition of wool's properties. Biodegradability and sustainability make wool suitable for some technical applications (Patnaik 2015). Technological applications of wool, at present, include building insulations (thermal and sound) (Schmidt et al. 2004), as well as the sorption of pollutants, mainly indoor air pollutants (Mansour 2018) and heavy metals (Hanzlíková et al. 2018). The usage of wool in technical applications can be boosted by applying chemical improvements, such as those used to enhance resistance to fire, microbes, and keratin consuming insects (Sunderland et al. 2014). The large scale and long tradition of the wool industry, as well as the appeal of the wool itself as a natural fiber, have long stimulated innovation in other fiber processing industries (Kuffner and Popescu 2012).

History of wool in the textile industry

Wool has been used since ancient era o as material to make various clothing and high-quality fabric in the modern textile industry (curtains, pillows, rugs, socks, blankets, carpets, and bedding) (Strand et al. 2010). Wool felts were known a long time ago in China, Babylon, and Egypt. Shearing of the wool, rather than pulling, was made possible by the invention of cutting tools in the Iron Age. The finest wool was Merino sheep wool, which was bred in Spain in the fourteenth century. At the end of the eighteenth century, sheep breeding began in Australia. Sheep can be found in any country in the world. Major wool-producing countries are Uruguay, New Zealand, China, South Africa, Argentina, and Australia (Eberle et al. 2004).

Classification of wool in textiles

There are many different types of sheep breeds, classified according to their wool into five basic types, which are Medium, Fine, Long, Coarse, and Crossbred (Macpherson 2012).

Wool has a widespread popularity as a garment fiber and for interior textiles. Nowadays, wool still a superb, sought-after apparel fiber with even more modern set of technical attributes. Likewise, for carpets and rugs, wool sets the benchmark of high quality and performance that could compare with other fibers (Johnson et al. 2003). Wool fiber for clothes provides clothing comfort and functions as a thermal insulator. Silky, combed yarns, and fine wool fibers are tightly constrained so they can scarcely crimp. Fine combed yarns enclose less air and therefore provide less insulation. Bulky woolen yarns have a looser structure, so that crimps which develop inside the fibers can entrap vast amount to offer proper protection against cold (Hossaini 2011).

The hygroscopicity of wool enables absorption of moisture up to a third of its weight without feeling wet, where liquid perspiration can chemically bind. Water droplets are repelled on the wool surface, but water vapor can be absorbed rapidly. However, liquid water is absorbed very slowly. Such behavior is called "hydrophobic." Wet wool dries very slowly (Van Emden and Carr 2010) Wool fiber is comfortable Next-to-skin. The softness of wool depends on its individual fiber diameter. Types of soft wool are Lambswool and fine Merino wool. Coarse fiber of about 30 μm

can irritate the skin (Millington and Rippon 2017). Wool cannot quickly burn, which makes it suitable for protective clothing.

Wool fiber in textile manufacture

Wool fiber in clothes manufacture

Wool is the main animal fiber used in textiles. The fibers are the hair of animals, most often sheep. Of course, this is a renewable resource (Lien Chen and Burns, 2006); as long as sheep feed on grass, they will continue to grow wool. Annually, at average, a sheep produces between 2.3 and 3.6 kg of raw wool which must be sheared (Fantilli, Sicardi, and Dotti 2017). Wool production from sheep usually depends on variations in the supply of nutrients to the follicles, which can exert a considerable influence on the rate of fiber production and the characteristics of the fleece. Increased feed intake can compensate for the effects of pregnancy and lactation. Production of wool with growth and age could be related to changing patterns of feed intake and diet selection, which affects sheep age and season of shearing. Wool production is influenced by the age and sex of animals, and by reproduction in the ewe. Less wool is grown by young animals per unit of feed intake, presumably due to competition for nutrients between follicles and other tissues. Maximum fleece weights in sheep have been observed as from three to 5 years of age. Weather could be a factor for wool production. Exposure of areas of skin to heat or cold, or experimentally induced localized changes in skin temperature, are indication of wool growth, especially length growth rate (Jamshed Khan et al. 2012). Wool is readily recycled because of biodegradable advantages of fiber, where the fiber can be returned to the land as a fertilizer, as it is rich in elements found such sulfur and nitrogen (Johnson et al. 2003). Wool has been a versatile fashion fiber used by many cloth brands and sportswear like Adidas, where the wool growers have demonstrated toward the fiber. Wool is the most profitable enterprise to run on countries trade (AWI 2018). The wool industry is a green industry which aims for the future development of the textile industry. Merino wool is one of the high grade materials used to produce high-quality textiles, and garments produced from this wool are preferred by markets and consumers. Also, wool has high demand in high-end fashion and is indispensable in the high-end products by well-known apparel and different fashion brands around the world. MCloskey (2019) reported that wool is used to manufacture worsted fabrics and high-end garments. The global textile industry is also researching for different product segments, and aspiring the move of using wool from traditional suits to new fashion markets and leading trends, such as the use of wool in casual wear, sportswear, and fashion accessories. Craig Vanderloef from Adidas company stated in 2018 that utilization of wool has much interest because it is the fabric and fiber that is often in line with our vision in Adidas running to deliver the best of performance and style to the consumer.

Hugo and Ross Turner (2016) reported that wool fibers are much better than manmade fibers due to its natural constituents which make them adaptable to the environment. Researchers recently found that under certain conditions, you can wear wool which is 20 – 25% lighter than with manmade garment and have the same insulation and comfort as well as hats and socks.

The Badland jackets are well-known outdoor clothes as they are wind and water-resistant. Neulana Protect fabric is a result of development by the Wool Development Center set up in 2013 by AWI and Chinese company Nanshan. The fabric is made from 100% Merino wool and without any chemical treatment with optimal technology. RÖJK Superwear makes wind and water-resistant shell made from chemical-free, super tightly woven and supple Merino wools. The experimenters were very enthusiastic about how well the Badland Jacket organized the body temperature while cross-country skiing and snow-shoeing. It is a great turning point for alternatives to softshells made from petroleum-based fabrics (RÖJK Superwear 2018).

The main benefit of increasing use of natural fiber such as wool instead of using manmade fiber for global textile market is because natural fibers have advantages of biodegrading in marine and

land environment, therefore do not lead to environmental issue such as microfiber pollution. This is an important difference between natural and manmade fibers where the last one needs to be accounted for Life Cycle Assessment to be credible and scientifically defensible (Henry, Laitala, and Klepp 2019).

Wool carpet

Wool products have longer average lifespan, and can be recycled or reused (Russell et al. 2016). Wool are advantageous as carpet due to its superior resilience and manifestation retention resistance to staining and great flame retardancy (Johnson et al. 2003). Carpets are mostly made by long and coarse wool. Wool carpets provide benefits that should be considered when choosing floor coverings. At home, the noise-reducing properties of carpets provide comfort, and at the office, they inaudibly increase job satisfaction and performance. Carpets control noise by reducing the floor noise, absorbing airborne noise and minimizing the transmission of noise through floors into other rooms (McNeil 2014). Wool carpets can absorb sound equally well as some dedicated acoustic materials, and reduce sound generation on, and propagation through, floors. Consequently, at home, the acoustic properties of carpets provide comfort (Ingham, McNeil, and Sunderland 2012), and natural ability to improve indoor air quality, inherent safety and sustainability, combined with their esthetics and prestige value to give the purchaser a justifiable feeling of satisfaction. McCall and McNeil (2007) found that wool carpets have the great ability of reducing the rates of nitrogen dioxide (NO₂) from buildings and furnishing materials (Spicer et al. 1989). Wool carpets have the potential for closed-loop recycling encompassing returning the used carpet to the soil as a fertilizer (Watson and S.A, 2005).

Global wool carpet production

The oldest wool carpet was found in Siberia, dating back to 500 BC, while the oldest wool textile was found in Denmark, dating back to 1500 BC. These reports prove that the existence of reliable sources of the fabric has changed greatly. Since then, wool has played a major role in the textile industry (Kumar and Suganya 2017). The carpets sector is the major end-use of strong wool, which makes up around 45% of global wool products. There is no doubt that wool will be widely used in carpets in the foreseeable future. There is still a large market for traditional hand-knotted rugs from wool, which should continue to be so. The production of wool tufted carpets in Europe peaked around 1990. There is, however, still a large global supply of strong crossbred wool (Ingham et al. 2016).

According to the report, the wool carpet market is expected to grow more vigorously which can influence the global economic structure with a higher revenue share. The wool carpet market holds the potential to impact its peers and parent wool carpet market, as the growth rate of the market is being accelerated by increasing product demand, growing disposable incomes, innovative products, raw material affluence, and changing consumption technologies.

A vast quantity (about 400 million kg) of wool is used in carpet manufacturing throughout the world annually, particularly in the traditional woven carpet industry. While globally Australia is considered as apparel wool growers, New Zealand produces excellent broader micron wool, which is suitable for carpet yarns. New Zealand produces 130 million kg of carpet grade wool, and is the largest supplier of carpet wool in the world, along with other countries such as UK (35 million kg), China (50 million kg), Commonwealth of Independent States (C.I.S) (10 million kg), India, Iran, and Turkey (70 million kg each), Argentina 5 million kg, others 125 million kg are also producers of carpet grade wool (Propa et al., 2018).

Wool socks

Table 1 shows the wool classification in textile industry. Organic production has become the gold standard of the sustainability movement and fashion products. As sustainability efforts have increased across the apparel and textile industries, consumers are being exposed to an increasing

variety of information and label claims. In the wool product market, the terms are appearing with greater frequency with the growth of consumer interest in sustainable fashion products just by selling sustainably produced socks (Teko 2011). Wool socks are animal fiber products with organic and alternative labeling schemes, which include eco-friendly, natural, and sustainable labels (Bernard et al., 2013).

Table 2 shows comparison of yarn parameters (twist coefficient, number of rubs at yarn break down, air permeability, and thermal conductivity), and some finishing processes that can affect the abrasion resistance of socks in detail between wool and PAC fiber. The abrasion tests were made on socks produced from wool and PAC fibers. It was found that the use of polyester acrylics increased the abrasion resistance of the socks. However, silicone softeners were found to degrade this property. The abrasion resistance of wool socks was significantly higher compared to PAC (Marmarali and Oglakcioglu, 2009).

Properties of wool

Among all textile fibers, wool has a very complex structure. This complex structure has characteristics of excellent flexibility, moisture absorption, flame resistance, warmth, coolness, odor absorption, biodegradability, recyclability, breathability, resilience softness, absorbs noise, safe, and easy to handle. Technical industrial applications of wool fiber, in reality, have some limitations due to cost. However, wool fiber has gained much more attention for technical utilizations in comparison with manmade fibers.

Wool fiber self-extinguishing

Wool fiber has been used to manufacture lightweight materials in aircrafts, ships, trains, and automobile upholstery, by using the Canesis LincSpun spinning technology that enables spinning of finer yams from wool much better than previously conventional technology. Wool is naturally advantageous for fire resistance. Wool has a relatively high LOI value of about 25 and a low flame temperature of about 680°C. Its high ignition temperature of 570–600°C is a consequence of its higher moisture regain of 8–16%, depending on relative humidity, high nitrogen (15–16%), sulfur (3–4%), and low hydrogen (6–7%) content by weight. While organo-sulfur compounds are generally flame retardant to some degree, the disulfide-containing cystine links are easily oxidizable, and this can offset some of the anticipated natural flame retardancy. Preoxidation of wool and hence cystine

Table 1. Wool classification in textile.

Type of Wool	Fine	Medium, Crossbred	Long, Coarse
Breed	Merino, Rambouillet	Southdown, Corriedale	Lincoln, Romney, Karakul
Diameter	Finest wools, 15–23 µm)	Medium fine, 24–30 µm	Coarse, over 30 µm
Length	50–120 mm	120–150 mm	Over 150 mm
Crimp,	Highly crimped	Normal crimp	Low crimp, straight
Origins	Australia, South Africa	Argentina, Uruguay	New Zealand, Great Britain
Utilization	Fine outerwear, knitted and woven, shawls, sock	Heavier, more robust, sporting clothing	Carpets,traditional furniture coverings

Table 2. Comparisons of samples knitted with 100% wool and 100% PAC yarn.

Materials	Yarn count /Nm	Twist coefficient	Number of rubs at yarn break down	Air permeability	Thermal conductivity (W/mK)
100% acrylic	16/1	104.9	4313	1786 mm/s.	0.031
100% wool	/401	112.2	1750	2460 mm/s	0.036

Table 3. Comparison of thermal transitions as flame-retardancy of different fibers.

Fiber	Tp, °C pyrolysis	Tc, °C ignition	LOI, %	Hc kJg ⁻¹
Wool	245	600	25	27
Cotton	350	350	18.4	19
Acrylic	290	>250	18.2	32
Polypropylene	470	550	18.6	44
Polyester	420–447	480	20–21	24

to cysteic acid residues restore this expected retardant activity. Oxidized wools can have greater inherent flame retardancy than other fibers. Table 3 shows the comparison of thermal transitions of the more commonly used fibers. By various chemical modifications, wool fibers can extend their flame-retardancy such fluorochemical treatment wherein enhance natural stain-resistance. (Johnson 2003). Sheep wool is a self-extinguish material that is potential for use in wide range of applications, such as seals for heat-generating appliances such as ovens, and clothes dryers (Owens and Burns 2004). As natural fiber, wool has a large potential for innovation in the new technical industry (Millington and Rippon 2017).

Wool thermal and sound insulator

Sheep wool began to be marketed and promoted as an alternative insulating material (thermal and sound) in building construction to improve the mechanical and ecological performances, as well as creates a link between textile and construction markets (Corcadden, Biggs, and Stiles 2014; Zach et al. 2012). Wool is also used as vibration insulator and cavity absorber (Helepiciuc 2017). The waste of wool which cannot be used for textile has been investigated as an insulating material (Patnaik et al., 2015) for replacement of commonly used insulation materials by using natural and renewable materials which have better environmental and primary energy values (Korjenic et al. 2015). Wool fiber has been used for manufacturing panels for sound absorption as the soft wool mat is 100% recyclable compared with polyester fiber (Bosia et al. 2015).

Wool is also used in civil engineering as thermal and acoustic insulations (facade and roof insulations) or in composites with different matrices (polymeric, earthen, or cementitious).

Symons, Clarke, and Peirce (1995) found that the application of natural/sheep wool shows better thermal insulation and sound absorption properties in comparison with using fiberglass and mineral wool. The result also indicated that bulk density of wool fibers is inversely proportional to air flow in pore structures of insulation panels, resulting in an increase of bulk density of wool fibers, leading to greater thermal insulating properties (Zach et al. 2012). Ballagh (1996) reported that wool is an excellent acoustic material and it gives maximum sound absorption at 170 mm thickness. Sheep wool panels have an absorption coefficient of about 0.84 at 2000 Hz which can be improved by up to 6 dB reduction of transmission loss on plasterboard walls. Wool can also be used as vibration insulator and cavity absorber, providing improvements of up to 10 dB. Corcadden, Biggs, and Stiles (2014) indicated that application of such waste wool offers a competitive material toward conventional insulation materials and adds benefits to economical and sustainable development of green buildings. An example of insulation material is soft mats. Soft mat is composed of 100% sheep wool with 4–6 cm thickness, mainly used for pitched roofs insulation. Semi-rigid panels are composed of 70–80% sheep wool and 20–30% polyester, and are 5–12 cm thick. They are applicable on walls due to the stiffness achieved through partial fusion of the polyester fibers.

Reinforcement materials

Fiber-reinforced composites are widely applied in various fields such as automotive, civil engineering, aerospace, sports equipment industries, etc. In comparison with metals and unreinforced plastic materials, animal fiber composites offer higher strength and flexibility, as well as corrosion

protection. The use of sheep wool as a filler of cement in order to produce mortar and plaster brings several advantages for environment and energy as reinforcement for cementitious or plastic matrices (Fiore et al., 2019), as well as cement mortar panels, instead of using synthetic ones such as asbestos in building materials due to its carcinogenic properties (Ardanuy et al. 2011). Materials from natural resources such as wool, cotton, or natural leather have been used as decorative surface materials for interior parts and seats since the beginning of industrial automobile production (Stevens 2010).

Das et al. (2017) investigated waste wool and biochar-based hybrid composites, and showed that wool is helpful in enhancing the limit oxygen index (loi) and some mechanical characteristics such as tensile and flexural strength. The charring ability of wool helped in a gradual reduction of heat release, while the use of composite led to less smoke release and reduced peak heat release rate, in comparison with virgin polypropylene composite. Waste wool is also used in making fiber-reinforced composite bricks, where incorporation of wool gives in an increase of 37% compression strength.

Štirmer, Milovanović, and Sokol (2014) reported that wool reinforced cement composite mortar provides better workability, small capillary action, better mechanical and compressive strength, as well as better thermal conductivity. Use of waste wool in mortar resulted in a decrease in density with increase in fibers volume, which makes it better material with less weight, which can be helpful in restoring historical buildings and ceilings.

Wool fertilizer

Sheep wool has been proposed as a raw wool constituent, particularly its protein hydrolyzates, for fertilizer application. Apart from waste which has been disposed, sheep wool waste as fertilizer nutrient sources for crop plants had been proven to be a good source of nitrogen (Vončina and Mihelič 2013). Wool blocks and granules are 100% natural and biodegradable materials commonly applied in soil for nutrients fertilization which has no negative impacts on the environment: no soil pollution, water contamination, nor soil degradation, besides maintaining biodiversity as well as preventing toxic substances to infiltrate groundwater or soil surface (Maria and Pacurar, 2015).

Wool geotextiles are used for the protection of drainage ditch bank and for steep slope in a gravel pit. Right after installation, the geotextiles wool provides immediate protection against water erosion and land sliding. The wool compounds release carbon, nitrogen, and oxygen after biodegradation, which enhances growth of protective vegetation (Broda et al. 2017).

Vončina and Mihelič (2013) studied sheep wool as a fertilizer without chemical treatment for asparagus production for 3 years. In the second year, use of sheep wool produced the highest content of soil NO₃-N, and resulted in the highest yield of asparagus. Maria (2015) and Ordiales et al. (2016) used raw wool in the form of pellets for soil quality improvement and fertilizer for the production of tomato and broccoli. The authors/researchers concluded that wool pellets are a viable option as alternative biofertilizer, as they resulted in more profitable yield of tomato and broccoli, in comparison with conventional fertilizers. The addition of unwashed and cut sheep wool in the field as a nutrient source for crops and soil amendments showed positive results with respect to plant growth and productivity of mangold and basil. The plants cultivated with sheep wool for soil amendment yielded 40-142% more.

Sheep wool hydrolyzate (a substance produced by hydrolysis) can improve growing conditions of plants, by increasing contents of essential elements N, C, and P in the soil (Vončina and Mihelič 2013). Applied hydrolyzed wool also acts as a chelating agent for microelements available from soil which are required for plant growth (Evangelou et al. 2008). Overall it is confirmed that waste wool has potential to be used as a fertilizer, but there is still need of a sustainable process to overcome all the existing problems and to consider its industrial scale application.

Catalytic photocatalytic

Wool is an abundant natural protein fiber that contains up to 95% by weight of pure keratin (Eslahi et al. 2013), and 18 types of amino acids (Aluigi et al. 2017) such as Glycine, Alanine, Serine, Naline, Threonine, Cystine, Lucien isomers, Aspartic acid Glutamic acid, Methionine, Histidine, Hydroxylsine, Phenylalanine, Arginin, Tyrosine, and Tryptophane, which is used widely in making textiles and apparel. Coarse or short wool fibers, usually treated as waste, if ground into fine powder, have been found to exhibit quite a number of functional applications in diverse fields. Wool has advantages to be catalytic and support catalytic. Two studies have clarified that wool can be used as primary catalytic, which are by McNeil, Sunderland, and Leighs (2017), who investigated wool as catalyst for nitroaldol reaction between aromatic and heteroaromatic aldehydes and nitromethane, to form beta-nitro alcohols, and by Häring et al. (2016) who proved that wool is an effective solid sulfonic acid catalyst.

Wool fiber has been used for chemical conversion of carbon dioxide (CO₂) through the use of catalytic converter. By combination with KI, wool can function as a catalyst in the industry, for cycloaddition coupling of CO₂ with propylene oxide (PO) to form propylene carbonate (PC) in a solvent-free condition. The advantage of wool fiber is it can be used for five cycles without apparently losing the activity. This catalysis system offers an environmentally friendly approach to recycle CO₂ into usable substances, and can be used to catalyze the coupling of other mono-substituted terminal epoxides with CO₂. Wool powder-KI to serve as an efficient, green catalyst system for the fixation of CO₂ (Chang et al. 2018).

Wool fiber had also been used as catalyzer in intensification process of Henry Reaction in a spinning mesh disc reactor, where copper was immobilized on wool to produce a superior catalyst to the homogeneous copper triflate system. The copper heterogenized on wool yielded higher batch conversion (85%) in the same timeframe. However, with reuse, it had been deactivated and, therefore, cannot be considered a stable catalyst. Overall, the results proved that copper immobilized wool in SMDR can be used to improve the conversions for the Henry reaction, which is promising for the SMDR to be extended for use to other solvent-based traditional response (Shivaprasad et al. 2017).

Wool water and air pollutants remover

Wool fiber had been investigated to treat wastewater by dyes removal, as wool has enlarged pores, where diffusion and penetration of dyes molecules within wool pores, result in increased sorption (Saleem, Pirzada, and Qadeer 2005). Oil spill pollution is one of the important issues and challenges worldwide, due to its environmental and economic impacts. Sheep wool fiber (Greener Cleaner) presents great potential as an inexpensive, easily available alternative adsorbent for oil removal. Besides, sheep wool is capable to collect and completely remove oil from the oil spill site (Sulyman, Namiesnik, and Gierak 2017). Wool has also been proven to remove diesel fuel, crude, base, vegetable, and motor oil (Radetic et al. 2008).

Wool fiber is an excellent sorbent material to remove heavy metals such as Ag and Cu from industrial stream and purify wastewater supplies. Wool contains 17 types of amino acids; one of them is cystine that reacts to form stable metal mercaptides. The mechanism of how wool metal interacts is by attribution to competition between hydrogen ions and metals for the same binding sites. For example, aspartic and glutamic acid act as the main binding sites for both silver and Copper (Freddi et al. 2001).

Sheep wool is among other materials that have low environmental impacts (Asdrubali, Schiavoni, and Horoshenkov 2012), which can reduce the environmental impact of operations on environment (i.e., energy consumption and CO₂ emissions) in several fields like construction. (Fiore, 2019). In comparison of environmental performances (transportation, usage, recycling, and release of greenhouse gases (CO₂, CH₄)) between three external and internal thermal wall insulation materials, which are mineral wool, calcium silicate and sheep wool, sheep wool has the lowest global warming potential (Klarić et al., 2015).

Wool fiber can sorb volatile and very volatile organic compounds (v/VOCs) from indoor air building materials such as formaldehyde Toluene, limonene, and dodecane, which improves work-place/environment quality and occupants health. Sheep wool fibers, with their keratinous chemical functionalities, provide platforms for sorption of v/VOC and known or suspected carcinogens substances (Mansour 2018).

Application of substances extracted from wool

Keratin

Wool contains up to 95% by weight of pure keratin (Li and Yang 2014). Keratin from wool is a tough, insoluble biomaterial that can perform fundamental structural role in many biological systems. In addition to possessing excellent physical properties, keratin is a biopolymer with high degree of chemical functionality, and exhibits many beneficial properties such as moisture sorption and thermal conductivity that synthetic polymers cannot achieve. Keratin is fibrous, three dimensional, and highly stable due to inter-molecular bonding of disulfide cystine 10–14% (Häring et al. 2016) amino acid and inter- and intra-molecular bonding of polar and nonpolar amino acids (Cardamone 2010).

Keratin is of significantly high interest in new product development for the pharmaceutical, medicinal, cosmetic, and biotechnology industries. Keratin materials extracted from wool can be fashioned into various shapes and forms of porous foams, sponges, mats, films, sheets, gels, microfibers, and bulk materials (Fernández 2019), as well as agricultural films and coatings for drugs (Eslahi, Dadashian, and Nejad 2013). These materials are being investigated for applications as bio-based, adaptable structures. Non-antigenic keratin has advantages for wound care, tissue reconstruction, cell seeding and diffusion, and drug delivery as topical or implantable biomaterial. As implantable film, sheet, or scaffold, keratin biomaterial can be absorbed by surrounding tissue to provide structural integrity within the body while maintaining stability under mechanical load, and in time can break down, to leave neo-tissue (Johnson et al. 2003; Van Dyke et al. 2002).

Medical materials which are absorbed (resorbed) by the body tissues after fulfilling their function are high-value products. Keratins can be processed into matrix that is considered biocompatible under vitro properties (Smiley et al. 2006). Therefore, they are well suited to the development of medical products with high-value, niche market applications (Yamauchi, Maniwa, and Mori 1998). Natural biopolymers have certain advantages in green chemistry including their pronounced role in preventing environmental pollution. Keratin application for the purification of metal contaminated natural and wastewater resources can be a promising technology (Khosa and Ullah 2013). Several metals such as mercury, copper, silver, cadmium, lead, chromium, and aluminum (Baek 2007). Nickel (II) chloride zinc (II) and Iron (II) ion had been evaluated for removal using keratin wool.

Wool has been used as absorbent by applying the keratin (keratin extracted from wool) to remove materials such as metal cations, contributed by functional groups suitable for metal binding. (Khosa and Ullah 2013).

Ceramides

Wool is a natural fiber that contains external lipids and a small amount of internal lipids (1.5%), which are of considerable interest because of their high proportion of ceramides. Internal wool lipids (IWLs), as shown in Figure 2, are rich in cholesterol, free fatty acids, cholesteryl sulfate, and ceramides (Schwan, Herrling, and Zahn 1986). The structure of the wool ceramide is presented in Figure 3.

Ceramides, which are the main component of the intercellular lipids in stratum corneum, or such as in human and animal hair, play an essential role in maintaining and structuring the lipid barrier, which affords protection against external insults and water loss through the skin (Coderch et al. 2003).

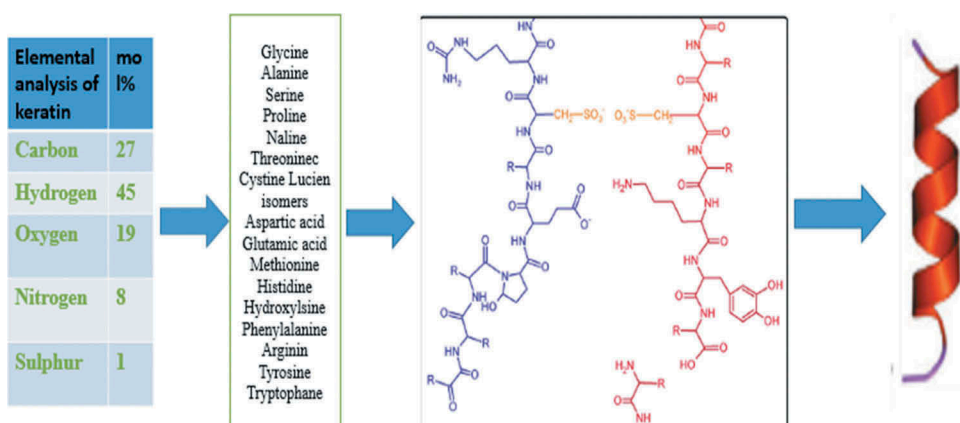


Figure 2. Chemical structure of keratine.

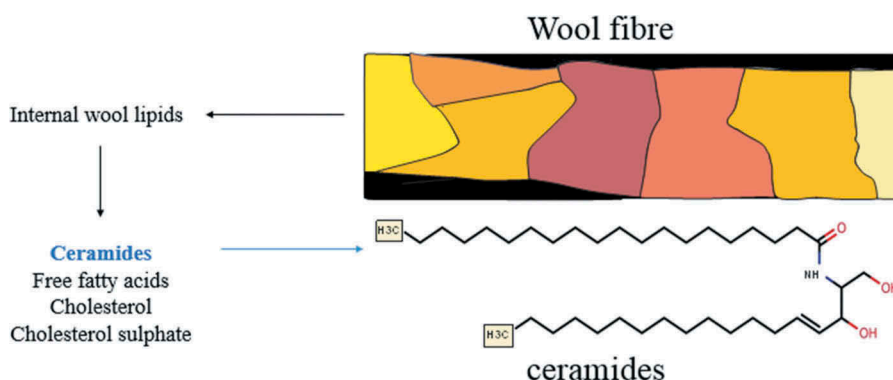


Figure 3. Wool ceramide.

Moreover, IWL iposomes have been proven able to improve the properties of human hair and skin when applied on them. IWL liposomes applied on skin can reinforce the skin barrier integrity and increase its water-holding capacity (De Pera et al. 2000).

Application of IWL liposomes to chemically treated hair samples can improve the mechanical strength properties of the fibers (Mendez et al., 2008). Therefore, IWL extracts structured as liposomes are suitable for incorporation into pharmaceutical or cosmetics formulations for the treatment and care of skin and human hair. Ceramides extracted from wool fibers is of special interest both to textile companies, given the added value by-product provided by a clean technology, and to cosmetic or dermatopharmaceutical companies, as sourcing of natural, novel, and beneficial extract for topical application (Ramírez et al. 2008).

In cosmetic usage, ceramide helps to hold skin by formation of protective layer that reduces the loss of moisture content and acts as protection against visible destruction from pollution. Ceramide also has been proven to confer better properties to human hair and skin (Ramírez et al. 2009), besides, as one of the main ingredients in anti-aging product (powerhouses) to support cuticles naturally.

The utilization of physiological lipids according to these parameters has potential as a new form of topical therapy for dermatoses. An alternative strategy to improving barrier function by topical application of the various mature lipid species is by enhancing the natural lipid-synthetic capability of the epidermis through the topical delivery of lipid precursors. Some ceramide supplementations are able to improve disturbed skin conditions (Luisa Coderch et al. 2003).

Lanoine (extracted wool wax)

Wool grease is separated from wool during its cleaning process, which involves the use of detergents for grease removal (Margenat et al. 2008). Wool wax differs from other waxes in terms of many of its chemical and physical properties which are soft and greasy, and tenacious and unctuous to the touch, as well as soluble in many organic solvents. It is unique in its high content of cholesterol, which has been used in commercial applications in many fields (Gillespie 1948). Products made from refined wool wax extracted via different methods such as soxhlet and supercritical fluids (such as lanolins derivatives) are the key ingredient in some ointments, beauty care products, pharmaceuticals, and some other substances such as waterproof coatings, hydrophobic vehicles (they are good vehicles for aqueous solutions, solids, and non-hydrolyzable drugs, vehicles for hydrolysable drugs such as white ointment, anhydrous lanolin, hydrophilic petrolatum, and emollients cleansing cream), and in the preparation of waterin- oil creams.

Lanolin by US Pharmacopoeia is purified wax-like substance obtained from the sheep wool waste, that has undergone some operations such as decoloration, and deodorization. The advantages of lanoline, along with melting point near normal human body temperature, make it applicable in some pharmacological treatments as well as medical treatment (Hanzlikova et al. 2015). Lanolin is a pale yellow, oily, grease substance with a faint properties odor, extracted and determined from raw wool. Textile processing properties are closely related to lanolin residues in the fiber (Alzaga et al. 1999). Lanoline, grease obtained from sheep wool, is a complex matrix composed principally of a mixture of esters involving more than 138 different aliphatic acids, combined with about 75 different alcohols (Jover, Moldovan, and Bayona 2002). Lanolin is widely used in soaps, and medical preparations. It is also used as an effective carrier for plant growth “promoting hormones” in the healing of tree wounds, and as a substitute for mineral oils in insecticidal sprays in the agricultural field. In the textile industries, wool wax is used to manufacture rayon, and the alcohols derived from the wax can be converted by sulfonation and neutralization into highly efficient wool scouring detergents equally effective in hard or soft water; a further product “wool oleine” is used as a lubricant in the carding and combing of wool fibers. Lanoline can be used in manufacturing of linoleum, in plastics, polishes, paper sizing, and compounding of rubber, as an egg preserver and in adhesives. (Gillespie 1948). Lanolin has outstanding advantages as a temporary rust preventer (Greene 1933). Each tonne of raw wool contains around 150 kg of lanolin, 40 kg of suint (soluble contaminants such as potassium salts from perspiration and feces), 150 kg of dirt (soil), 20 kg of vegetable matter, and residues of insecticides; leaving 640 kg of wool fiber (Bhavsar 2020).

Peptone

Among microbial culture media, Peptone is one of the most expensive lab materials. Sheep wool is used to produce microbial peptone by chemical process or enzymatic methods. Wool peptone is very rich in protein (67.8 g per 100 g) and ash (29.2 g per 100 g) contents. The most abundant amino acid, Glutamic, is found in wool peptone with the content of 8175 mg per 100 g. Wool also has high content of cystine (5042 mg per 100 g). The optimal concentration of wool peptone in microbial growth is specified as 5 g for the fungi and 6 g for the bacteria. Wool peptone can be one of the solutions to reduce the total price of microbial culture media, besides being an environmentally friendly method for removing these residues (Taskin et al. 2016).

Peptones are protein hydrolyzates that are prepared from proteinaceous material. (Keratin is nutrition for microorganisms). They contain secondary protein derivatives such as polypeptides, dipeptides, and amino acids. They provide a readily assimilable source of nitrogen, which is water-soluble, does not coagulate upon heating, and is therefore particularly suitable for inclusion in microbiological culture media (AL-Bahri, AL-Naimi, and Ahammed 2009).

Wool peptone medium is capable to boost bacteria growth such as *Penicillium chrysogenum*, and *Bacillus subtilis* to moderate levels. In a previous study *Aspergillus niger* fungi biomass

production was achieved, weighing 8.17 g. Second best growth performance for *Escherichia coli* was achieved with WP among the tested peptones. (Taskin et al. 2016). The mechanism of wool fiber hydrolysis by microorganisms suggested by the American scientist E. Race represents a sequence of transformations: proteins – peptones–polypeptides–water + ammonia + carboxylic acids (Pekhtasheva et al. 2011).

Summary and recommendation

This paper has reviewed wide breadth of research on wool and its utilizations as natural fiber. More exciting applications of wool fiber have been designated and applied in different fields, where some substances extracted from wool fiber such as lanoline, keratin, ceramides, and peptone have been investigated for utilization in cosmetics, pharmacotherapy, and medical area. Besides textile use, wool fiber is used in manufacturing of building materials (wool fiber as thermal and sound insulation) and in agriculture sector, where wool waste is recovered to be fertilizer. Wool fiber is a versatile natural material for use as adsorbent for the removal of wastes from industrial waste effluents, and filtering indoor air. Wool fiber from sheep is abundant and inexpensive, thus is potential as alternative to replace synthetic wool, along with potentials for researchers to seek other uses of wool fiber.

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